← Go Back
Hardware

Portenta Vision Shield

Tutorials

Blob Detection with Portenta and OpenMV
Saving Bitmap Camera Images to the SD Card
Connect Portenta Vision Shield to TTN with LoRa® Technology
Creating a Basic Face Filter With OpenMV
Training a Custom Machine Learning Model for Portenta H7
Using the Portenta Vision Shield - Ethernet with Arduino IDE
Using the Portenta Vision Shield - Ethernet with OpenMV
Getting Started With the Portenta Vision Shield Camera
Voice Commands With The Arduino Speech Recognition Engine
Connecting to The Things Network Using OpenMV
Portenta Vision Shield User Manual

Home / Hardware / Portenta Vision Shield / **Voice Commands With The Arduino Speech Recognition Engine**

Voice Commands With The Arduino Speech Recognition Engine

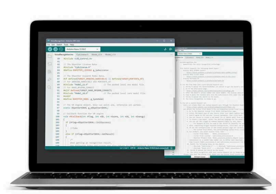
Control your device with voice commands using the Arduino Speech Recognition Engine

Author · Pablo Marquínez · Last revision · 05/27/2025

What Is Speech Recognition

Speech recognition is a technology field that captures, interprets, and computes a voice to transform it into text (TTS). Once the voice has been transformed into text, it can be applied to different applications, from speech dictation, to command-voice controllers, health monitoring, robotics and artificial intelligence or accessibility, among many others.

The Arduino Speech Recognition Engine



Help

The Arduino Speech Recognition Engine, which is powered by Cyberon, is one of the best platforms available on the market to perform speech recognition on embedded microcontrollers. The full compatibility with multiple Arduino boards and with the Arduino IDE makes it very flexible, efficient, and easy to use, both in hobby projects and in professional products.

The Arduino Speech Recognition Engine can be further customized based on the license type, check the **Licensing section** of this tutorial to learn more.

Overview

This tutorial shows how to use the Arduino Speech Recognition Engine to configure multiple voice commands and perform custom tasks based on the recognized commands. To do that, it is necessary to perform a series of steps to register your board and activate your trial and free-of-charge license. Afterward, you will test the demo sketch and learn how to create your own custom voice commands.

In case you are interested in unlocking the full potential of the tool, the tutorial will teach you how to acquire a paid license to unblock the free license limitations.

Goals

Set the voice command

- Get your board serial number and the required files to set up the library
- Get the license and use the demo sketch
- Create a new project with your own custom voice commands
- Learn how to acquire and use a paid license in case you want to remove the free license limitations

Required Hardware and Software

To use the Arduino Speech Recognition Engine, you will need one of the following boards:

- **Arduino Portenta H7 (any variant)** + Portenta Vision Shield (**LoRa** or **Ethernet**)
- **Arduino Nicla Vision**
- **Arduino Nano 33 BLE Sense Rev 1**
- **Arduino Nano 33 BLE Sense Rev 2**
- **Arduino Nano RP2040**

In the case of the Portenta H7, remember that is always possible to add an external microphone as additional hardware, instead of using the Portenta Vision Shield.

Additionally, you will need the following hardware and software:

- **USB-C® cable** (x1)
- **Arduino IDE 1.8.10+**, **Arduino IDE 2.0+**, **Arduino Cloud Editor**, or **Arduino-cli**

Instructions

The Arduino Speech Recognition Engine is a solution powered by Cyberon that requires a series of additional steps to unleash its potential. This tutorial will explain how to:

- Get the Serial Number of your Arduino device
- Install the library on your IDE
- Test the free demo sketch

In case you would like to extend the engine functionalities, you will have to purchase a voucher on the Arduino Store and follow the next steps:

- Fill in the required information on Cyberon's website
- Get the required files and activate your license

Setup

Setup the Library

There are three libraries, you will need to install one or another depending on which board you are using:

- **Portenta H7 Family**: Cyberon_DSpotterSDK_Maker_PortentaH7
- **Nicla Vision**: Cyberon_DSpotterSDK_Maker_NiclaVision
- **Nano 33 BLE Sense (Rev1 & Rev2)**: Cyberon_DSpotterSDK_Maker_33BLE
- **Nano RP2040**: Cyberon_DSpotterSDK_Maker_RP2040

Inside each of the libraries and under the folder "extra", you will find additional documentation made by Cyberon. Check them out in case you need more information

Go to the Library Manager, search for the library that you need for your board and install it.

In case you need more instructions about how to install libraries, read this guide.

Get The Serial Number

To use the Arduino Speech Recognition Engine, you will need a free trial license or paid license. In any of the cases, the serial number of the board that you are using is necessary to activate the license.

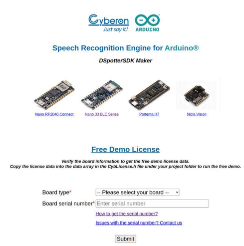
To get your board's serial number, and once you have the library downloaded, navigate to **File > Examples > Cyberon_DSpotterSDK > GetSerialNumber**.

Connect your board to the computer, upload the sketch to it and, once is done, open the **Serial Monitor** to see your device's Serial Number.

On the Arduino IDE 1.6.x or previous versions, you can also find the serial number as follow: select the board's serial port and click on `tools > Get Board Info`, you will see the "SN" number, save it for later.

Once you have the Serial number, open: https://tool.cyberon.com.tw/ArduinoDSpotterAuth/FDMain.php

- Select your board
- Fill in the "Board serial number" field
- Click the **Submit** button



Get License Page

Once everything is ready, click on the **submit** button to get your license, it will print an array of numbers for the license. **Save it in a safe place**, it will be used in the next step.

Test the Free Demo Sketch

1. Open the sketch **File > Example > Cyberon_DSpotterSDK > VoiceRecognition**
2. Navigate to the `CybLicense.h` tab.
3. Paste your license between the brackets, like in the following example:

```
#include <stdint.h>

const uint32_t g_lp
    0x00000000, 0x0
    0x00000000, 0x0
    0x00000000, 0x0
    0x00000000, 0x0
    0x00000000, 0x0
    0x00000000, 0x0
    };
```

Remember to replace the `g_lpdwLicense` values with your license ones.

Now switch back to the `VoiceRecognition` tab and upload the sketch.

You can now open the Serial Monitor where you can see the available commands that the CyberonSDK recognizes.

Serial Monitor Trigger and Command list

Now it is time to test the engine. Feel free to say out loud the commands and see how it recognizes your voice!

Please note that the sketch and license that you are using are a "demo" version with the following limitations:

- A maximum of 50 recognitions per each board start
- A 20 seconds delay before each trigger recognition

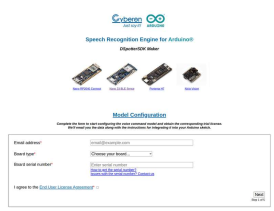
Customized Commands

To expand the features and human interaction of your projects, you can create your own voice commands with the Arduino Speech Recognition Engine. In this section, you will learn about how to use the Cyberon Model Configuration webpage to create a new project with custom voice commands.

These steps are compatible with the trial license, used in this tutorial, and the paid license

Go to Cyberon Model Configuration and fill in the required fields:

- E-mail address
- Board model
- Serial Number of your board
- EULA Agreement - Please, read it carefully



Cyberon Model Configuration

Click next, you will see a new page to:

- Create a new project
- Import an existing project

Create a New Project

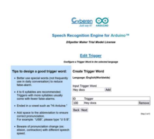
Warning: each project is bound to a single Arduino board with the declared Serial Number. If you would like to use the Arduino Speech Recognition Engine on another Arduino board, you need to create a new project from scratch and assign it to the new Serial Number.

To create a new project first you need to select the desired language for the speech recognition. Once is set, click **Create**.

Cyberon, New Project

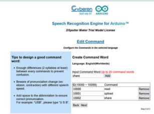
Now you need to configure the following steps to create a new trigger:

- Create the **Input Trigger word**, for example "Hey Arduino". The **Input Trigger word** will trigger the device to let the board know that you are going to say a command after that.



Cyberon Adding the Input trigger

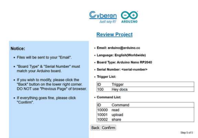
- Add the **Command** list. These commands will be used to do tasks on your sketch. For example, if you have a command which is `share`, later you will be able to get that command and proceed with some job inside the sketch easily.



Cyberon Adding the Command list

The next step is just to confirm that the data written is correct. Click on **Next** to finish.

On the next page you will see all the configurations already set. Check it out to confirm that everything is correct. In case something is wrong, click on **Back** to fix it.



Cyberon finishing model configuration

Once everything is checked, click **Confirm** and you will get the model header file (`model.h`). This header is part of your new project and you need to copy it onto your sketch folder to use it.

Warning: when a project is created and confirmed, it cannot be further modified.

You will now get some files in your e-mail inbox. Download them to your computer.

On the IDE, open the example **File > Examples > Cyberon_DSpotterSDK > VoiceRecognition** and click **File > Save As...** and type a name for your sketch.

Once it is saved, open your File Explorer, and navigate to your sketch path.

On your sketch directory, paste the files you have got in your e-mail inbox before:

- `CybLicense_<id>.h`
- `Model_<id>.h`

Now to implement the **Input Trigger Command** and the **Command List** open the sketch and change the following headers with the downloaded ones:

```
...

#include "CybLicense.h"

...

#include "Model_L1.h" ->

...

#include "Model_L0.h" ->
```

At this point, the project is set to be used. Upload the sketch and open the Serial Monitor. Now you can test your new **Input Trigger Word** and the **Command** list that you have created. Pronounce the new trigger words out loud, you will see the recognized words on the **Serial Monitor**.

Modify the Example Codes

If you want to execute custom actions with the **trigger** and **commands** you defined, you can do it inside the `VRCallback` function.

First you need to know your trigger and commands IDs. You can find them listed in your Serial Monitor, in

Keyword	ID
Hey Arduino	100
read	10000
upload	10001
share	10002

Here is the `VRCallback` function modified so you can easily give functionality to the commands detection:

```
// Callback function fo
void VRCallback(int nFl
{
  if (nFlag==DSpotterSD
  {
      //ToDo
  }
  else if (nFlag==DSpot
  {
      /*
      When getting an r
      the following inc
          nID         Th
          nScore      nS
                      Th
          nSG         nS
                      Th
          nEnergy     nE
                      Th
      */
      //ToDo
      switch(nID)
      {
          case 100:
            Serial.prir
            // Add your
            break;
          case 10000:
```

We added a `switch...case` that evaluates the `nID` variable and compares it with the different IDs. Inside each case add your custom code to be executed with the trigger or commands

```
...
        switch(nID)
        {
            case 100:
              Serial.print
              // Add your
              break;
            case 10000:
              Serial.print
              // Add your
              break;
            case 10001:
              Serial.print
              // Add your
              break;
            case 10002:
              Serial.print
              // Add your
              break;
            default:
              break;
        }
...
```

Unlock Limitations (License)

In case you want to further customize your model or add additional commands or trigger words, we have the right licenses for you:

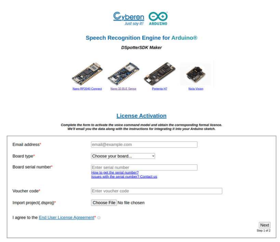
- **Speech Recognition license.** Vouchers available on the Arduino Store.
- **Pro License.** Feel free to contact us to unlock all the limitations.

There are different licensing options to fit the needs of your project. Read more in the Licensing section (Voucher codes). Note that you need to have an already existing project to use a paid license. Check the previous section to know more.

Browse to Cyberon's Licensed Project Configuration.

Fill in the required fields:

- E-mail address
- Board
- Serial Number of your board
- Voucher code
- Import project: the `.dsproj` file that you received in your e-mail inbox during the creation of the project.
- EULA Agreement - Please read it carefully



Cyberon Model Configuration with a voucher code

Click next, review your project options and press continue.

You will get a new e-mail with the new License and Model headers.

> Warning: in case you are using a Speech Recognition License (not Pro), please note that deployed models are not customizable. You can always purchase an additional voucher on Arduino Store to generate a new model for the same hardware.

Open the sketch you have duplicated in the Create New Project section.

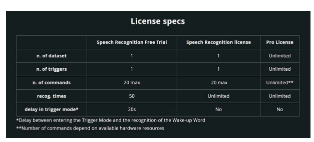
Repeat the replacement of

```
...

#include "CybLicense_<id

...

#include "Model_<id>.h"

...

#include "CybLicense_<id
```

Upload the sketch to your board, and you will have unlocked the limitations of the trial version.

Licensing

Here you can know more about the different licenses and limitations of each device. Remember that the code that you get for your license is also called a **voucher code**.



License specs

	Speech Recognition Free Trial	Speech Recognition license	Pro License
n. of dataset	1	1	Unlimited
n. of triggers	1	1	Unlimited
n. of commands	20 max	20 max	Unlimited**
recog. times	50	Unlimited	Unlimited
delay in trigger mode*	20s	No	No

*Delay between entering the Trigger Mode and the recognition of the Wake-up Word

**Number of commands depend on available hardware resources

License options for the Arduino Speech Recognition Engine

Free Demo

- Compatible with the CDSpotterSDK library
- Number of SET: x1
- Number of TRIGGERS: x1
- Number of COMMANDS: 4
- Configure TRIGGER: Fixed (not configurable)
- Configure COMMANDS: Fixed

- Delay in Trigger mode: no
- Hardware binding: no

Speech Recognition Free Trial

Once the number of recognition is reached the model will stop working and you will need to manually reboot the target.

- Compatible with the CDSpotterSDK library
- Number of SET: x1
- Number of TRIGGERS: x1
- Number of COMMANDS: 20 maximum
- Configure TRIGGER: Configurable
- Configure COMMANDS: Configurable
- Recognition times: x50
- Delay in Trigger mode: 20 seconds [1]
- Hardware binding: Yes

[1] Please note that the delay in Trigger mode means a 20 seconds delay between entering the trigger mode and the recognition of the wake-up word.

Speech Recognition License

- Compatible with the CDSpotterSDK library
- Number of SET: x1
- Number of TRIGGERS: x1
- Number of COMMANDS: 20 maximum
- Configure TRIGGER: Configurable
- Configure COMMANDS: Configurable
- Recognition times: Unlimited
- Delay in Trigger mode: No delay

PRO

- Compatible with the CDSpotterSDK PRO library
- Number of SET: Unlimited (depends on the hardware)
- Number of TRIGGERS: Unlimited (depends on the hardware)
- Number of COMMANDS: Unlimited (depends on the hardware)
- Configure TRIGGER: Configurable
- Configure COMMANDS: Configurable
- Recognition times: Unlimited
- Delay in Trigger mode: No delay
- Hardware binding: Yes

Next Steps

After getting the demo sketch working, we encourage you to start implementing this on your own project to unleash the full potential of the Arduino Speech Recognition Engine.

In case you are looking for more information, you can check out Cyberon's documentation:

- Inside each of the libraries and under the folder "extra", you will find additional documentation made by Cyberon, as well as the `Readme.md` file that contains the requirements and the steps shown in this tutorial
- Cyberon's youtube channel contains a series of step-by-step tutorials that can help you to know more about how to use the Arduino Speech

You can also consult **Cyberon's official Maker User Guide**

Conclusion

In this tutorial, you have learned what is the Arduino Speech Recognition Engine, the different licenses and limitations, and how to acquire one. Once you have your license, you learned how to test it with the free demo sketch and how to create a project with your own custom voice triggers. Eventually, you learned how to upgrade a free already-made project to be used with a more advanced license.

Suggest changes

The content on docs.arduino.cc is facilitated through a public GitHub repository. If you see anything wrong, you can edit this page here.

Need support?

Help Center
Ask the Arduino Forum
Discover Arduino Discord

License

Was this article helpful?